Callable Bermudan Cross-Currency Swap - Quantitative Model Summary

Status: generic quantitative specification
Model: two Hull-White rate factors + lognormal FX + Bermudan LSM
Numeraire: domestic money-market account; values reported in domestic currency

1. Product and notation

Consider a two-currency swap with domestic (`DOM`) and foreign (`FOR`) cash-flow legs, optional initial/final notional exchanges, and exercise dates $t_1 < ... < t_M$. Let

- $S(t)$ be FX in units of `DOM` per one unit of `FOR`;
- $P_d(t,T)$ be the domestic collateral discount bond;
- $P_f*(t,T)$ be the foreign effective discount bond consistent with the domestic collateral convention and cross-currency basis; and
- positive cash flows denote receipts by the exercise holder.

A foreign cash flow $CF_f(T)$ has pathwise domestic value $S(T)\ CF_f(T)$. All cash flows are discounted with the domestic numeraire. For a cancellation right decided at t_i, partition value into:

- A_i: cash flows common to exercise and continuation, including surviving flows through a delayed effective date;
- J_i: exercise-only settlement value at t_i; and
- Y_i: cancellable future cash flows and later optionality if the trade continues.

The holder exercises when

$$J_i > E^{Qd}[Y_i \mid F(t_i)] + epsilon\_ex$$

and the decision-date value is $A_i + \max(J_i, C_i)$, where C_i is the conditional continuation estimate. For a zero-fee cancellation, $J_i = 0$: the holder cancels when continuation is negative. For an option to enter the swap, replace J_i by the state-dependent PV of the entered swap plus settlement.

2. Joint stochastic model

Work under the domestic risk-neutral measure Q_d. Each currency has a Hull-White one-factor short rate and FX is lognormal. With S = `DOM/FOR`:

$$dr_d = [theta\_d(t) - a\_d\ r\_d]\ dt + sigma\_d(t)\ dW\_d$$

$$dr_f = [theta\_f(t) - a\_f\ r\_f - rho\_fS\ sigma\_f(t)\ sigma\_S(t)]\ dt + sigma\_f(t)\ dW\_f$$

$$dS/S = [r\_d - r\_f]\ dt + sigma\_S(t)\ dW\_S$$

$theta\_d$ fits the domestic curve under Q_d. Calibrate the foreign Hull-White curve fit under its foreign/effective numeraire measure, then transform its dynamics to Q_d; the displayed negative quanto drift is that measure change for the stated quotation. Equivalently, verify the domestic-measure identity $S(0)\ P_f*(0,T) = E^{Qd}[D_d(0,T)\ S(T)]$. Changing numeraire or reversing the FX quotation changes the adjustment and requires a fresh derivation.

The instantaneous Brownian correlation matrix is

$$R = \begin{bmatrix} 1 & rho\_df & rho\_dS \\ rho\_df & 1 & rho\_fS \\ rho\_dS & rho\_fS & 1 \end{bmatrix}$$

R must be symmetric and positive semidefinite. A constant matrix is the base model; piecewise-constant correlation is an extension. FX is Black-Scholes only in its diffusion form: stochastic rates make the full hybrid model non-analytic.

Multi-curve assumption

The base model has one stochastic rate factor per currency and deterministic forecast/discount basis. If P_c^F is a forecast pseudo-discount curve and P_c^D the modeled discount curve, freeze

$$B_c(0,T) = P_c^F(0,T) / P_c^D(0,T)$$
$$P_c^F(t,T) = P_c^D(t,T)\ B_c(0,T) / B_c(0,t).$$

Forward coupons are derived from $P_c^F(t,T)$. Stochastic cross-currency or tenor basis requires another factor and is outside this model.

3. Calibration

- **Curves:** bootstrap domestic collateral and foreign effective curves from liquid deposits/OIS/basis instruments. Enforce `F_FX(0,T) = S(0) P_f*(0,T) / P_d(0,T)` and reprice calibration instruments.
- **Domestic HW:** calibrate to domestic normal-vol swaptions on the exercise diagonal. Fix or tightly constrain `a_d`; regularize/bound piecewise `sigma_d(t)` to avoid an unstable joint fit.
- **Foreign HW:** apply the analogous foreign swaption fit and identification constraints for `a_f` and `sigma_f(t)`.
- **FX diffusion:** fit piecewise `sigma_S(t)` to selected ATM/benchmark-tenor FX vanillas in the full stochastic-rate hybrid. If several strikes are used, minimize weighted price error and report smile residuals; exact smile fitting needs a local/stochastic-vol extension.
- **Correlation:** use governed historical/implied estimates, validate PSD, and report independent stresses to all three correlations.

Calibration output includes fitted parameters, market/model price per helper, relative error, optimizer status, and curve/vol/correlation snapshot identifiers. Parameter stability and out-of-sample swaption/FX-option repricing are part of model validation.

4. Monte Carlo discretization

The event grid contains all fixing, accrual, payment, notional-exchange, decision, effective, and settlement dates, plus intermediate steps capped at a chosen maximum interval. Simulate correlated normal increments using a Cholesky/eigen factorization of `R`.

- Use exact conditional Gaussian transitions for Hull-White factors when available.
- Integrate domestic short rates consistently to obtain pathwise discount factors.
- Update log FX with `d log S = [r_d-r_f-0.5 sigma_S(t)^2]dt + sigma_S(t)dW_S`, using a discretization consistent with the rate integrals.
- Value conditional zero-coupon bonds analytically from Hull-White states where possible; apply deterministic basis mapping for forecast curves.
- Use antithetics initially; validate Sobol/Brownian-bridge variants separately.

An independent non-callable XCCY cash-flow PV is the primary control variate and deterministic benchmark.

5. Bermudan exercise by least-squares Monte Carlo

Use a two-pass Longstaff-Schwartz algorithm.

Training pass. Simulate training paths and work backward from `t_M`. At `t_i`, the regression target is the value, discounted to `t_i`, of cancellable cash flows in `(t_i,t_{i+1}]` plus the optimized value at `t_{i+1}`. Exclude common `A_i` cash flows from both alternatives; add them outside the exercise comparison. Regress this target on state observed at `t_i`. A generic standardized basis is

```
1, r_d, r_f, log(S), PV_d, S*PV_f,
r_d^2, r_f^2, log(S)^2, r_d*r_f, r_d*log(S), r_f*log(S).
```

Use all alive paths for a cancellation right unless a validated candidate filter is specified. Record rank, condition number, coefficients, path count, and residual diagnostics. Freeze preprocessing and coefficients after training.

Pricing pass. Apply the frozen policy forward on independent paths. The first exercise stops only the cancellable tail; retain already-paid and delayed/common cash flows and add the exercise settlement. Discount every realized cash flow to time zero and average. This is the LSM lower-bound estimate. Re-train across seeds and evaluate each policy on one common frozen pricing sample to separate policy dispersion from ordinary pricing noise.

6. Measures, risks, and validation

Core outputs are callable NPV, non-callable NPV, embedded option value, Monte Carlo standard error/confidence interval, exercise/no-exercise probabilities, expected exercise time, per-leg PV, calibration diagnostics, and regression diagnostics. From the holder perspective:

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embedded option value = callable NPV - non-callable NPV >= 0
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up to covariance-aware Monte Carlo comparison error.

Risk is bump-and-revalue with common random numbers: domestic/foreign curve DV01, forecast-basis and cross-currency-basis risk, domestic/foreign swaption vega, FX delta/gamma, FX vega, three correlation sensitivities, and Hull-White parameter stresses. State whether each bump recalibrates the model and retrains the stopping policy.

Minimum validation set:

1. no exercise dates reproduce the non-callable XCCY PV;
2. zero-vol limits reproduce deterministic discounted cash flows;
3. identical currencies and unit FX reduce to a single-currency result;
4. one exercise date matches an independent European/nested-MC benchmark;
5. domestic-numeraire-discounted domestic assets and discounted FX-converted foreign assets are martingales under `Q_d`;
6. prices converge across path counts, grids, bases, and independent training seeds;
7. representative multi-exercise cases compare the LSM lower bound with a dual upper bound or trusted nested-MC benchmark; and
8. all differences use paired pathwise errors under common random numbers, or the covariance-aware error of the difference.

References

- F. A. Longstaff and E. S. Schwartz (2001), "Valuing American Options by Simulation: A Simple Least-Squares Approach."
- QuantLib reference documentation: https://www.quantlib.org/docs.shtml